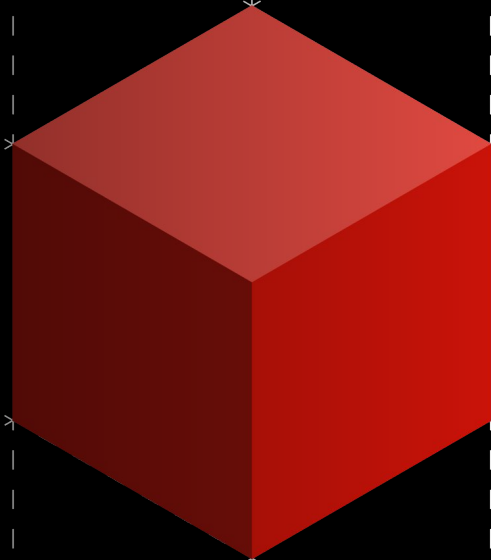




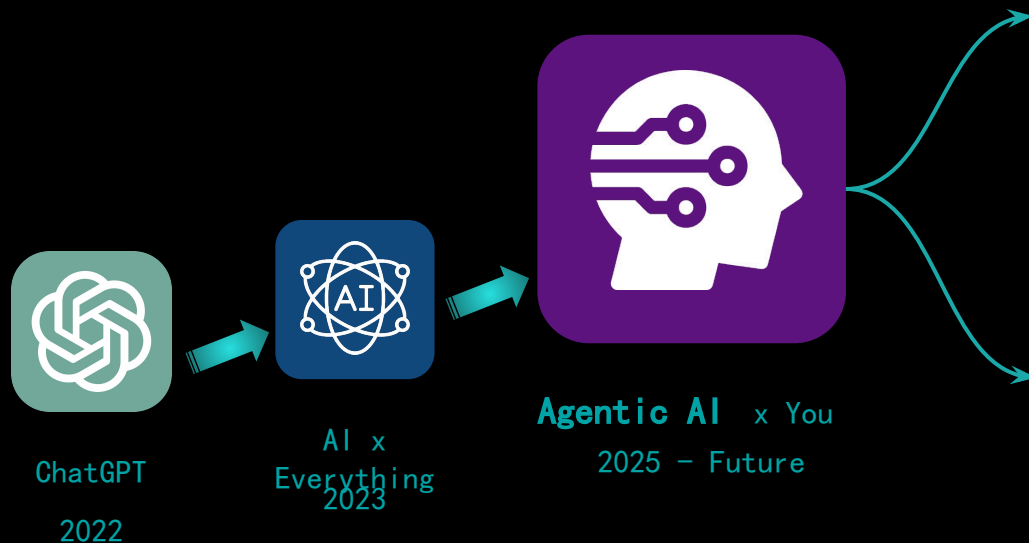
TiDB: The Data Foundation for the AI Era

Powering an ecosystem of autonomously generated and dynamically evolving intelligent applications

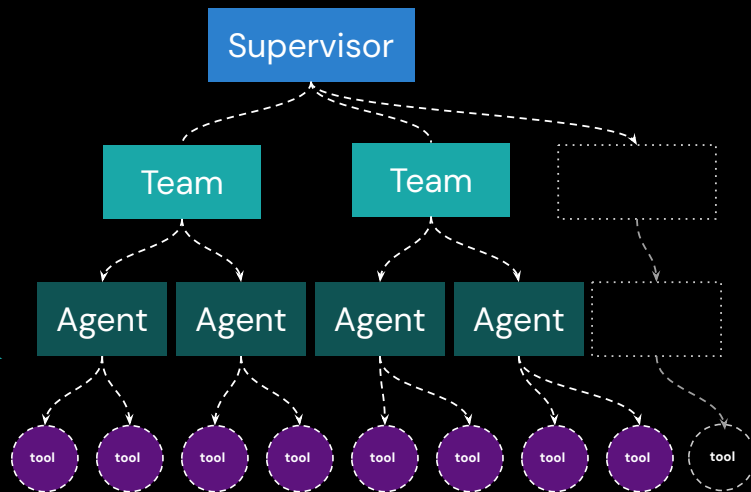


AI Evolution Demands Stronger Data Infrastructure

New Age in AI – Agentic AI



Autonomous Agentic RAG



Apps Evolve: from Human-Made to AI-Created

	 Human-shaped	 AI Agent-shaped
Concurrency Model	Concurrency bounded by human thinking time; fan-out ≈ 1	Near-zero thinking time; agents trigger agents $\rightarrow$ fan-out > 1 ; QPS $\approx$ active_agents $\times$ request_rate $\times$ fan-out
Schema & Metadata	Deterministic human-designed schema & queries	Millions of heterogeneous, AI-generated schemas & queries
Governance & Tuning	Manual, coarse governance — per-user / app limits, offline budgets, static guardrails, human tuning.	Automated, fine-grained governance — per-agent quotas and budgets, automatic guardrails, self-tuning.
Data Type	Occasional multi-model	Routine hybrid: OLTP + vector + full-text + light analytics in one flow

Determinism



Non-determinism

The Evolution of Data Infrastructure



Downsides of Infrastructure Limitations

1

Inability to handle dynamic schemas and multimodal queries

2

Slow elasticity, unable to keep pace with AI agents' short life cycles

3

Lack of native support for vector and semantic search

4

Cost structure unsuitable for large-scale AI workloads

TiDB Empowers AI Applications



TiDB: An All-in-one Data Foundation for AI

Challenge	Description	<u>Why TiDB?</u>
Data Model Diversity	Agents must reason across structured (SQL), semi-structured (JSON), and unstructured (vector, graph) data in real-time.	• “Whatever the agent generates, it runs” → Federated optimizer unifies OLTP + vector + full-text via pluggable column-index engines.
Schema Flexibility	Agents need a strong capability to support fast, flexible, consistent and ONLINE schema operations.	• Manage millions of evolving tables across tenants/agents — a capability forged responding to “1M–10M tables” for real-world asks • Handle rapid schema changes with online DDL and atomic versions does not slow down queries.
Unlimited Scalability	So many Agents working at the same time with millions of QPS/TPS/Tables, PB level data, and batch processing in 100TB, will be a HUGE challenges to databases.	• Elastic OLTP at very high QPS and data volume. • Built to stay fast under surges and fan-outs.
Mixed Workload	Constant interleaving of high-throughput transaction (state, memory), full-text search, light analytics, vector search and complex knowledge retrieval.	• Per-agent RU limits & real-time budgets to bound cost, enforce fairness, and isolate noisy agents. • Future self-tuning queries with RU / telemetry feedback so agents can optimize and adapt quotas dynamically.
Cost Visibility	Spreading Agents across multiple database clusters causes significant resource inefficiency and inflated management costs.	• Consolidate all Agents into one cluster will be impossible for traditional databases.



Introducing TiDB

The Distributed SQL Database Powered by PingCAP

Founded in
2015

100%
open source
software

4,000+
enterprise
adopters

Customers
across
25
countries



38K+ GitHub Stars



10K+ Adopters



800+ Contributors



5K+ Slack Users

Dify



manus



LobbyAI



Rengage

crowdSnap

TiDB Cloud Architecture

• SQL Layer

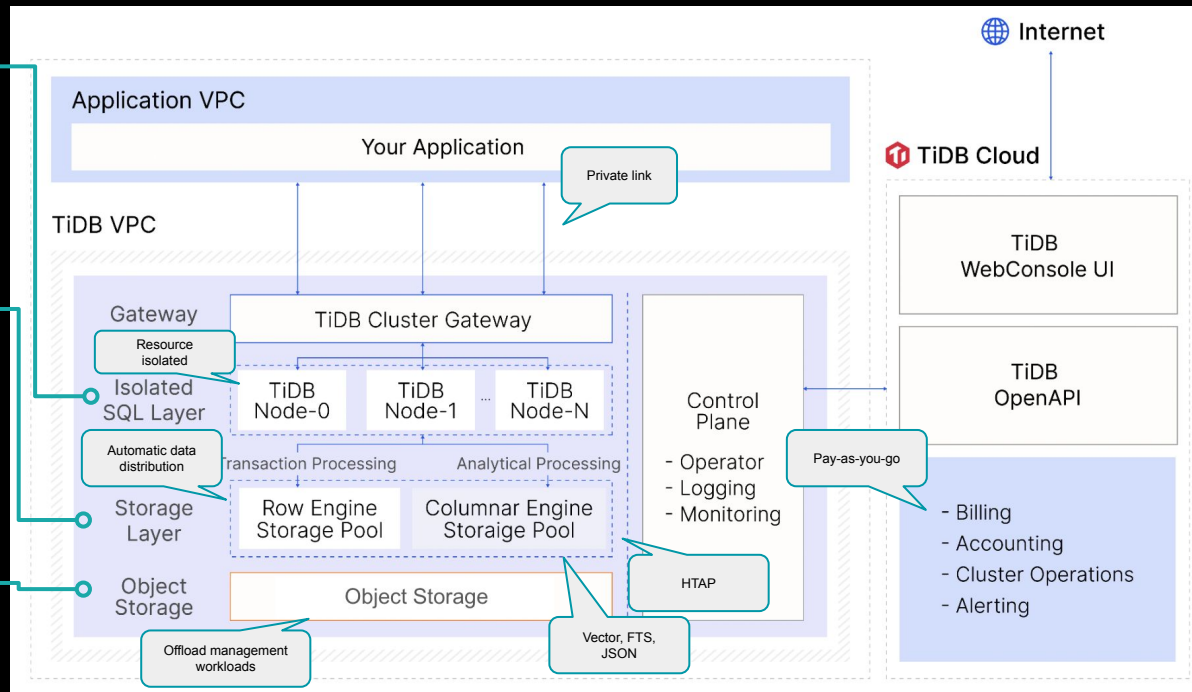
A stateless SQL layer compatible with MySQL that decouples compute from storage and supports on-demand scaling. Each node in this layer can handle both read and write requests.

• Storage Layer

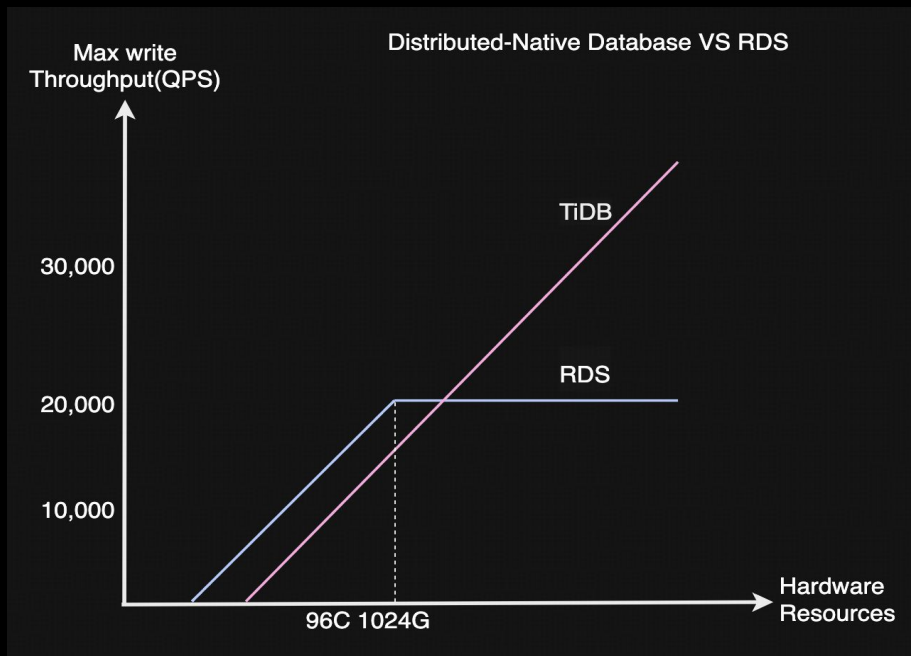
A distributed storage layer that supports both row-based and column-based storage engines. It provides built-in high availability and strong ACID consistency and can automatically scale to hundreds of nodes and petabyte-level data volumes.

• Object Storage Layer

Fully integrates the advantages of object storage—virtually unlimited capacity, infinite throughput, and low cost. It supports hot and cold data tiering, decouples background management operations from runtime processes, and significantly improves stability, cost efficiency, and resource scheduling.



Extreme Scalability – Remove Growth Bottlenecks



Traditional relational databases suffer from **single-node write bottlenecks**, whereas TiDB offers **linear scalability**.

Data Volume:

PB+

Batch Processing:

100TB+

QPS & TPS:

Millions

Number of Tables:

Millions

Backup & Restore:

PB+

Online DDL:

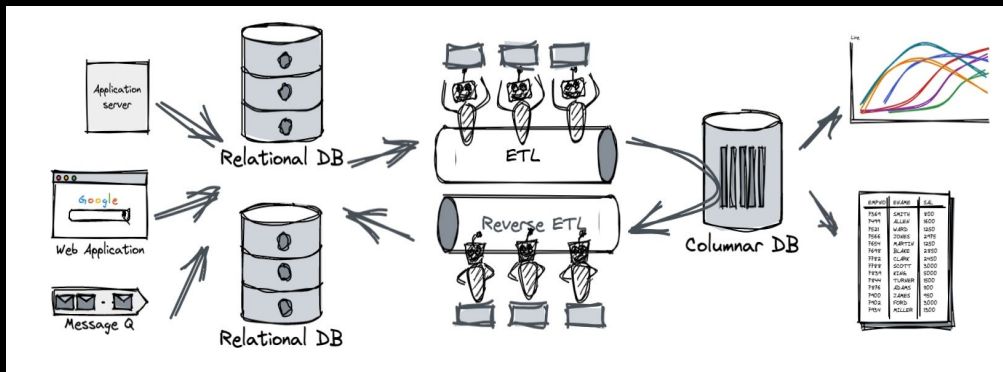


Resources Isolation:



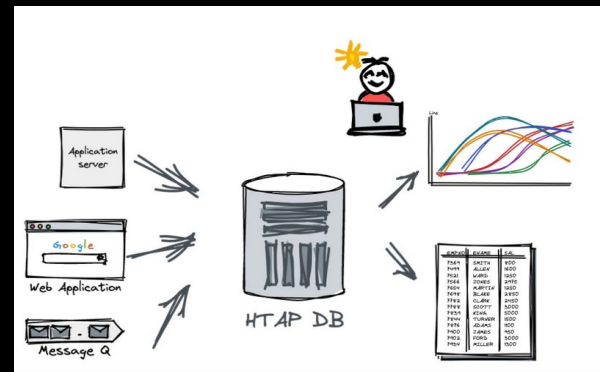
HTAP – Simplify Tech Stack

Designed for modern data-intensive applications



Might take a team several months

- Applications alone can't meet business growth needs—rely on data warehouses for support
- Cross-system, cross-team data development increases communication and maintenance costs
- Data timeliness is usually measured in days—hard to support real-time decisions



Only two engineers, one week of effort

- Elastic and HTAP-capable, supports rapid business growth
- Faster iteration, less operations, simplified data infrastructure
- Gain insights when the business happens

Online DDL – Empowering Continuous AI Innovation

Values to AI:

- **Seamless Schema Evolution:** Adapt data structures instantly as AI models and agents evolve.
- **Zero Disruption:** Keep real-time AI pipelines and applications running without downtime.
- **Agile Experimentation:** Support rapid iteration for model updates and feature engineering.
- **Always-On Infrastructure:** Ensure uninterrupted learning, inference, and data feedback loops.

	MySQL 8.0	TiDB
Add/Remove Fields	Table lock	No lock , DDL duration is in seconds.
Modifying field default values	Table lock	No lock , DDL duration is in seconds.
Modifying data precision	Table lock	No lock , partial lossless operation is completed in seconds.
Creating an index	Table lock	No lock , DDL duration is directly proportional to table data.
DDL Impact	Data delays in the primary-secondary replication.	There is no data delay impact on primary-secondary replication.

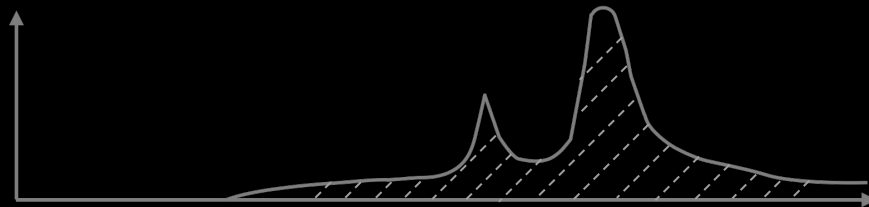
Elastic Scalability – Auto-adapt to Changing Workloads

The next-generation architecture brings exceptional elastic scalability to TiDB. TiDB **allocates resources on demand** based on real-time workloads, significantly reducing database costs.

- **Elastic Scalability**

- Automatic scaling in and out
- No compute resource allocation during idle periods
- Dynamic resource allocation during traffic spikes
- Resource reclamation during low-traffic periods

Workload – Throughput



Auto Resource Scaling

- **Pay-as-you-go Model**

DB Branching – Power Rapid AI Agent Iteration

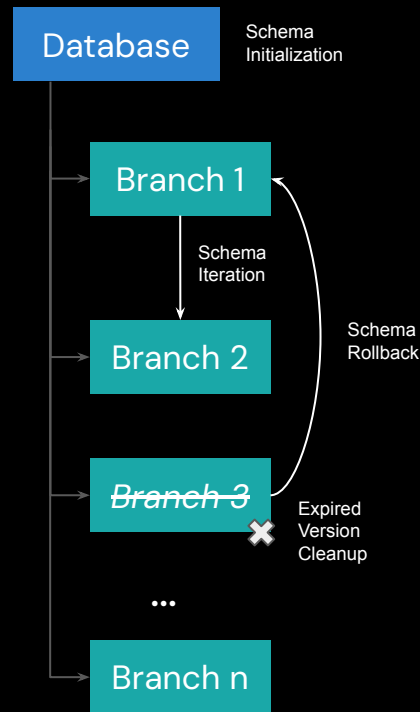
TiDB Branching enables AI companies to quickly create isolated data environments for each model version, agent, or experiment, enabling parallel iteration and secure isolation.

Capabilities

- **Fully Isolated**: Dev/Test Branch Environments
- **Fast readiness**: Copy-on-Write Technology
- **Automation Integration**: CLI & GitHub

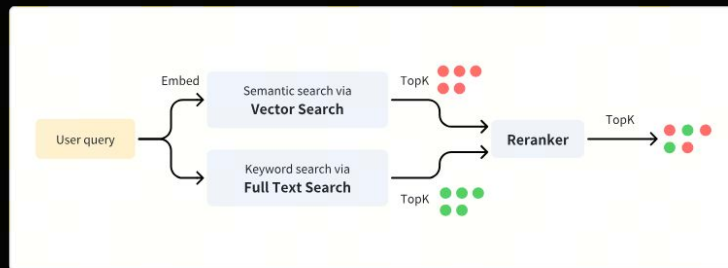
Advantages

- **Instant Startup**: Branches are created instantly, integrating smoothly into development pipelines.
- **Data Isolation**: Ensures production database safety and consistency.
- **Automated Integration**: GitHub triggers → auto branch creation, updates, and cleanup.
- **Easy Management**: Supports policy configuration via tidbcloud.yml and full CLI control.



Hybrid Search – Make AI Search Smarter

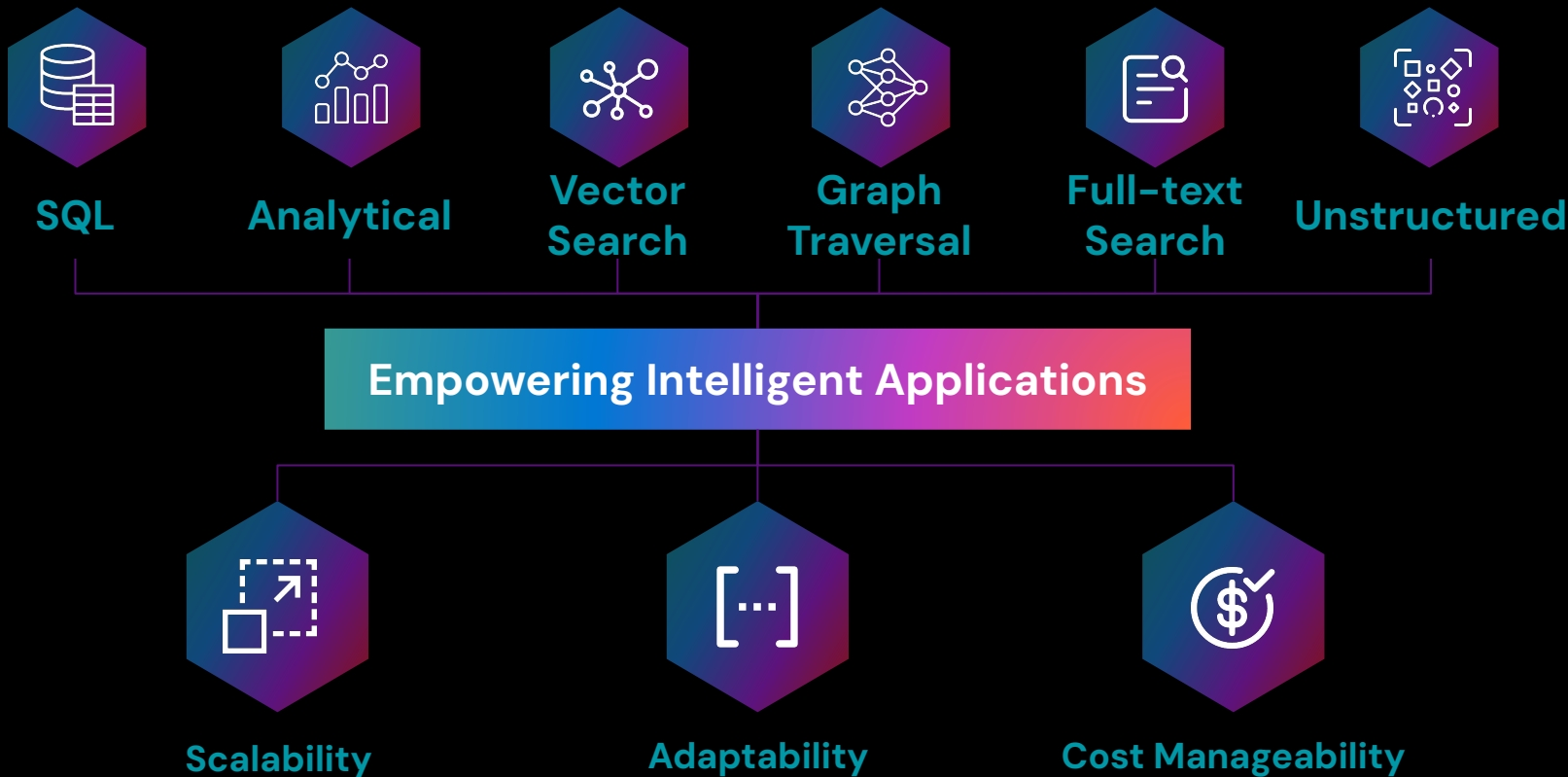
TiDB integrates structured data, vector data, document search, and analytical capabilities — providing a unified data foundation and Hybrid Search capabilities for AI applications.



Typical Use Cases

- **RAG (Retrieval-Augmented Generation)**
Unified storage of text and vectors to enhance the quality and consistency of LLM responses.
- **Semantic & Multimodal Search**
Enables semantic retrieval across languages and media types.
- **Recommendation Systems**
Builds personalized recommendation models through similarity computation.
- **AI Data Platform**
Integrates knowledge bases, business data, and embeddings to simplify architecture and operations.

One TiDB for AI Native Apps





Real-World AI Use Cases, Built on TiDB

Build intelligent AI applications — from chat agents and search assistants to knowledge graphs and real-time decision engines.



Retrieval-Augmented Generation (RAG)

Enable LLMs to generate more accurate, grounded answers by combining vector, full-text, and structured SQL queries.

[Learn more](#)



Knowledge Graph QA

Use multi-hop traversal and relationship-aware retrieval to power graph-driven question answering.

[Learn more](#)



Intelligent Analytics & Insights

Combine real-time analytical queries with semantic search to surface instant insights across massive operational datasets.

[Learn more](#)



Personalized Search & Recommendations

Blend structured filters, semantic vectors, and user signals to drive real-time relevance.

[Learn more](#)

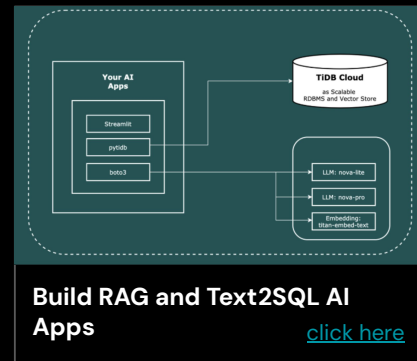
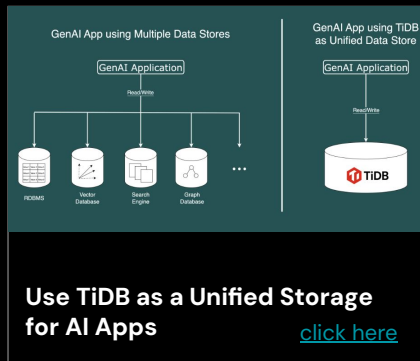


More Resources: AI Hands-on Labs

Before using TiDB to build your AI applications, you can try the following hands-on labs:

Build Vector Search Apps Using Jupyter Notebook [click here](#)

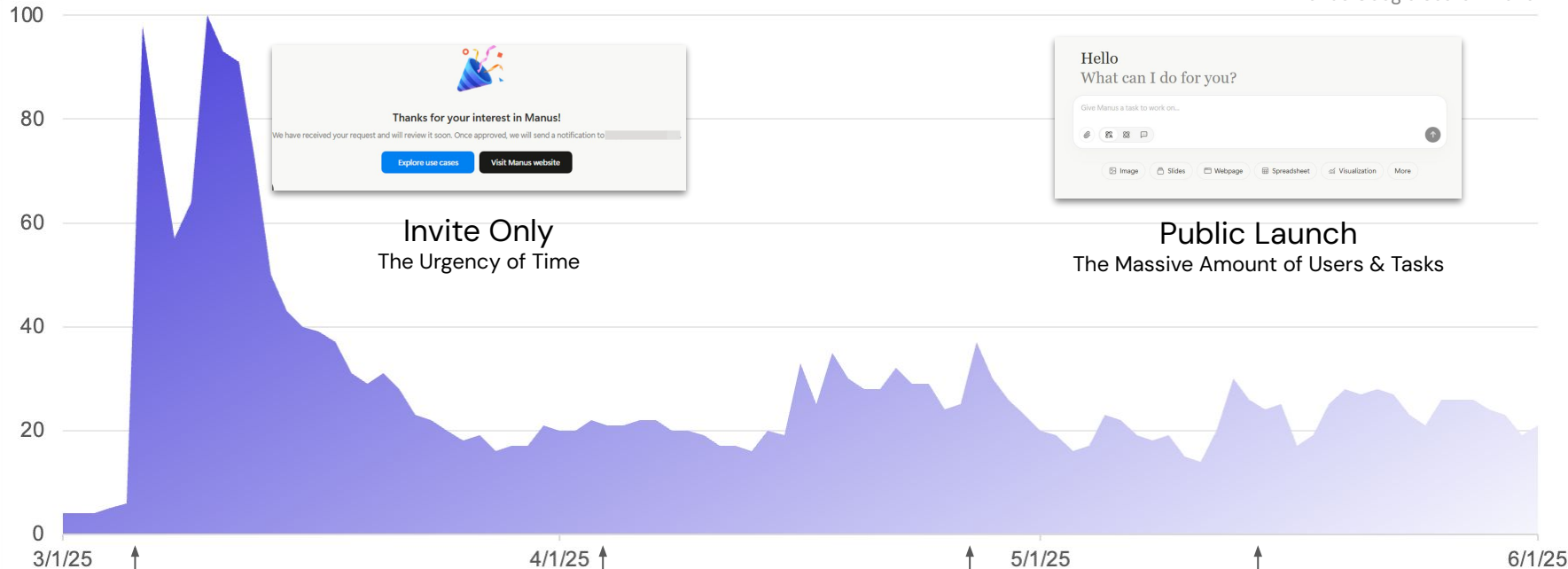
Build GraphRAG and Tool Calling AI Agent Using TiDB [click here](#)



TiDB: The Data Backbone Driving AI Customer Success

Manus: The world's First Truly Generic AI Agent

Manus Google Search Trend



891,877 views Mar 5, 2025
Manus is a general AI agent

Feel the future at <https://manus.im>



First Contact & Scoping



Go Prod



Exponential Data Growth

Manus: TiDB Meets the Key Requirements

- **Scalability**

Scale to millions of database instances:

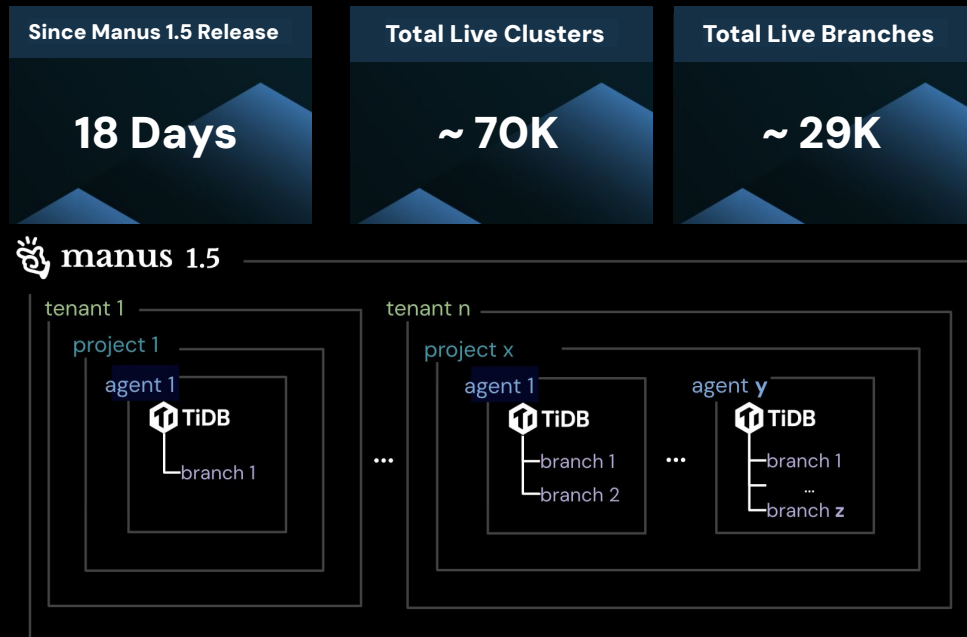
N tenants * X projects * Y agents * Z branches

- **Adaptability**

Per-agent/attempt/app schema updates without impacting operation or performance. Support **OLTP**, **OLAP** and **semantic search** workloads in one system

- **Cost Manageability**

Pay-as-you-go pricing perfectly matches tenants' intermittent database usage. Costs are transparent and can be allocated to each agent.



Manus agent mapping to TiDB cluster

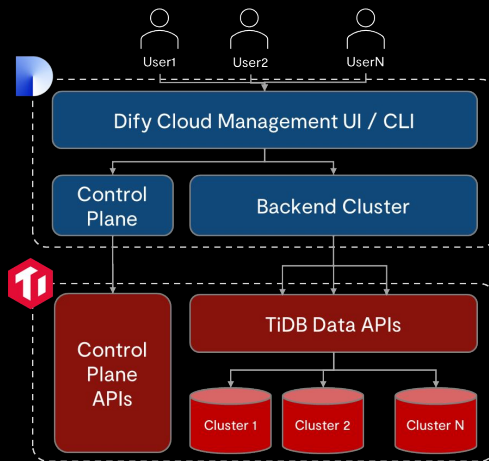
Dify.AI: SaaS Transformation with TiDB Hybrid Search

Dify.AI, the most popular LLM tool on GitHub with 114,000+ stars, has transformed their data architecture using TiDB Cloud to solve a critical challenge in GenAI platform development. By consolidating an extensive number of database containers into a unified system



Challenges:

- Managing 300,000+ databases led to **complexity and high costs**.
- Needed a **unified architecture** for multi-tenancy AI SaaS.



TiDB Values:

- 4.6x Faster Development** – Features in days instead of weeks.
- 60% Cost Reduction** – Eliminated over-provisioning.
- 80% Less Overhead** – Automated database management.
- Improved Performance** – Consistent query speeds for AI workloads.

Rengage.AI: Powering Autonomous, Real-Time Marketing at Scale

Rengage leverages AI Agents and TiDB to deliver large-scale, autonomous, and real-time personalized marketing.



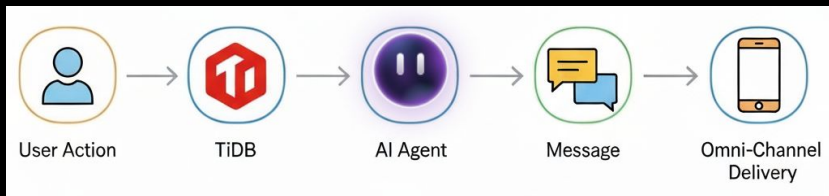
Challenges:

- Static journeys couldn't support real-time, individualized engagement.
- Scaling personalization caused latency, complexity, and high costs.
- Separate hot/cold data pipelines led to inefficiency and query delays.



TiDB Values:

- **High throughput & horizontal scalability** for massive marketing workloads.
- **Real-time data access** enables AI Agents to act instantly.
- **True large-scale personalization** powered by unified analytics and live data.



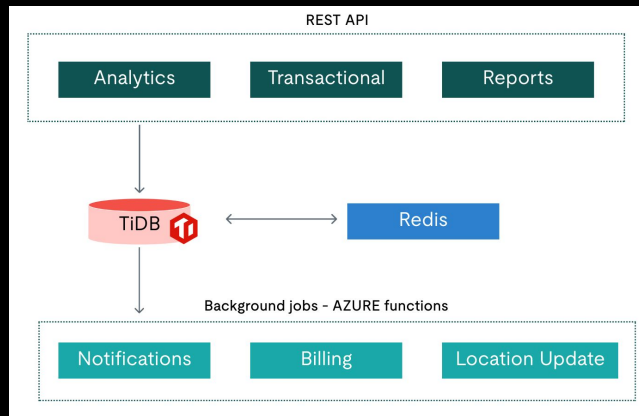
CrowdSnap Transforms Real-Time Analytics with TiDB

CrowdSnap, a Sri Lanka-based analytics platform, helps governments and event organizers analyze visitor data in real time. They started with a typical three-tier setup using MySQL, Node.js ETL jobs, and dashboards built on aggregated tables.



Challenges:

- Batch ETL caused data delays and prevented real-time analytics. T+1->15-60 mins.
- Maintaining multiple layers (MySQL, ETL, dashboards) increased complexity and cost.
- Scaling under heavy event workloads was difficult.



TiDB Values:

- TiDB unified transactional and analytical workloads in one platform.
- Real-time dashboards run directly on live data—no ETL or aggregates needed.
- Simplified architecture reduced maintenance and improved scalability.

Get started today!



Get
started



<https://tidbcloud.com/>



<https://pingcap.github.io/ai>



<https://labs.tidb.io>